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D15C / 55

EXPERIMENT NO.1

Aim: Implement Linear and Logistic Regression on real-world datasets

1. Dataset Source

The dataset used is the **Air Quality Dataset** from Kaggle, which contains real-world air pollution and meteorological data collected from urban sensors.

- **Source Link:** https://www.kaggle.com/datasets/fedesoriano/air-quality-data-set

2. Dataset Description

The dataset contains measurements of atmospheric pollutants and environmental parameters.

- **Features Used:**
  - T → Temperature (°C)
  - RH → Relative Humidity (%)
  - AH → Absolute Humidity
- **Target Variables:**
  - **Linear Regression (Continuous):** NO2(GT) – Nitrogen Dioxide concentration
  - **Logistic Regression (Binary, created):** air_quality
    - 1 → Good air quality
    - 0 → Poor air quality
- **Dataset Size:** Approx. 9,400 records (after preprocessing)
- **Characteristics:** Real-world urban environment data with some missing and invalid values (-200), which were cleaned via mean imputation.

3. Mathematical Formulation of the Algorithms

Linear Regression

Linear Regression predicts a continuous target variable:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_n x_n + \varepsilon$$

- (y) → predicted NO_2 concentration
- (x_1, x_2, ..., x_n) → features (T, RH, AH)

- (\beta_0) → intercept
- (\beta_1, ..., \beta_n) → coefficients
- (\varepsilon) → error term

Logistic Regression

Logistic Regression predicts a binary outcome using the sigmoid function:

$$P(y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)}}$$

- Probability is converted to class labels using a threshold (0.5)
- (1) → Good air quality, (0) → Poor air quality

4. Algorithm Limitations

Linear Regression Limitations

- Assumes a linear relationship between features and target
- Sensitive to outliers
- Cannot model non-linear or complex interactions

Logistic Regression Limitations

- Assumes a linear decision boundary
- Performance may degrade on highly imbalanced datasets
- Not suitable for highly non-linear classification without feature engineering

5. Methodology / Workflow

Step-by-step workflow:

1. Load dataset from Kaggle.
2. Remove empty columns.
3. Replace invalid values (-200) with NaN and fill missing numeric values using mean.
4. Select features for Linear Regression (T, RH, AH) and target (NO2(GT)).
5. Create binary target air_quality for Logistic Regression based on the median.
6. Split data into training (80%) and testing (20%) sets.
7. Train Linear Regression and Logistic Regression models.
8. Predict on test sets and evaluate performance.

6. Performance Analysis

Linear Regression Results

```
=====================================
LINEAR REGRESSION RESULTS
=====================================
Mean Squared Error (MSE): 1768.9854376887422
R2 Score: 0.06912668259409727

Linear Regression: Actual vs Predicted (First 5 Values)
   Actual_NO2  Predicted_NO2
0   58.000000     128.048653
1  115.000000     110.621287
2  160.000000     111.341586
3  113.091251     113.003885
4   51.000000     104.927017
```

Interpretation: Environmental features partially explain NO_2 concentration. R^2 is low, indicating other factors also affect NO_2.

Logistic Regression Results

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=====================================
LOGISTIC REGRESSION RESULTS
=====================================
Accuracy: 0.6490765171503958
Confusion Matrix:
 [[243 468]
 [197 987]]
Classification Report:
              precision    recall  f1-score   support

           0       0.55      0.34      0.42       711
           1       0.68      0.83      0.75      1184

    accuracy                           0.65      1895
   macro avg       0.62      0.59      0.59      1895
weighted avg       0.63      0.65      0.63      1895
```

```
Logistic Regression: Actual vs Predicted (First 10 Values)
   Actual_Air_Quality  Predicted_Air_Quality
0                   1                      0
1                   0                      1
2                   0                      1
3                   1                      1
4                   1                      1
5                   0                      1
6                   0                      1
7                   1                      1
8                   1                      0
9                   0                      0
```

Interpretation: The model correctly identifies good air quality better than poor air quality.

7. Hyperparameter Tuning

- **Linear Regression:** Closed-form solution; no hyperparameter tuning needed.
- **Logistic Regression:** Increased max_iter=1000 to ensure convergence. Regularization (C) can be tuned for better performance, but default was sufficient for this dataset.

Conclusion:

- Linear Regression was able to model NO_2 concentrations but had a low R^2 due to missing explanatory variables.
- Logistic Regression successfully classified air quality with moderate accuracy (64.9%).
- This practical demonstrates real-world application of regression algorithms for environmental datasets, including data preprocessing, feature selection, and model evaluation.